Smallben Wiring Reference

NodeMCU-32S (30-pin) • TP4056 • MT3608 • SD Card • RC522 RFID • I2S DAC/Amp • Controls • LEDs

1. Part Identification Quick Reference

ESP32 Board (30-pin NodeMCU-32S)

Left side (top→bottom): 3V3, GND, D15, D2, D4, RX2 (GPIO16), TX2, D5, D18, D19, D21, RX0, TX0, D22, D23
Right side (top→bottom): VIN, GND, D13, D12, D14, D27, D26, D25, D33, D32, D35, D34, VN, VP, EN

Slide Switch (3-pin)

Center pin = Common. Sliding the lever connects the center pin to either the left or right outer pin.

Potentiometer (3-pin, knob pointing away from you)

Pin 1 = Left outer leg | Pin 2 = Middle leg / Wiper | Pin 3 = Right outer leg

LED Legs

Long leg = Anode (+) | Short leg = Cathode (–), also marked by a flat edge on the plastic rim.

2. Power System & Charger (TP4056 + MT3608 Booster)

From	Pin	To	Target Pin	Notes
Battery	Positive (+) Red	TP4056	B+	Li-ion 3.7V
Battery	Negative (–) Black	TP4056	B–	Protected input
TP4056	OUT–	MT3608	VIN–	Direct connection
TP4056	OUT+	Slide Switch	Pin 1 (left pin)	Main power switch
Slide Switch	Pin 2 (Center)	MT3608	VIN+	Turns system ON/OFF
MT3608	VOUT+ (5.0V)	ESP32	VIN (top-right pin)	Set MT3608 to 5.0V FIRST
MT3608	VOUT+ (5.0V)	SD Module	VCC / 5V	Powers SD card
MT3608	VOUT+ (5.0V)	I2S DAC/Amp	VIN / VCC	Powers audio amp
MT3608	VOUT– (GND)	ESP32	GND	Common ground
MT3608	VOUT– (GND)	SD Module	GND	Common ground
MT3608	VOUT– (GND)	I2S DAC/Amp	GND	Common ground

■ CRITICAL: Set the MT3608 output to exactly 5.0V with a multimeter BEFORE connecting it to the ESP32, SD module, or amp. Overvoltage from an unset booster can destroy these components.

3. SD Card Module (SPI Bus 1)

SD Card Pin	ESP32 Pin Label	GPIO	Code Reference
VCC	VOUT+ on MT3608	—	5V power from booster
GND	GND	GND	Common ground

SD Card Pin	ESP32 Pin Label	GPIO	Code Reference
CS	D5	GPIO 5	SD_CS_PIN
SCK	D18	GPIO 18	SD_SCK_PIN
MISO	D19	GPIO 19	SD_MISO_PIN
MOSI	D23	GPIO 23	SD_MOSI_PIN

4. RFID-RC522 Module (SPI Bus 2)

■ The RC522 MUST be powered from 3V3 (3.3V) only. Connecting it to 5V will burn it out.

RFID Pin	ESP32 Pin Label	GPIO	Code Reference
3.3V	3V3	—	3.3V power from ESP32
GND	GND	GND	Common ground
RST	D22	GPIO 22	RFID_RST_PIN
SDA (SS)	D21	GPIO 21	RFID_SS_PIN
SCK	D14	GPIO 14	RFID_SCK_PIN
MISO	D35	GPIO 35	RFID_MISO_PIN
MOSI	D13	GPIO 13	RFID_MOSI_PIN

5. Audio DAC / Amplifier (MAX98357A / PCM5102)

I2S Amp Pin	ESP32 Pin Label	GPIO	Code Reference
VIN / VCC	VOUT+ on MT3608	—	5V power
GND	GND	GND	Common ground
BCLK	D27	GPIO 27	I2S_BCLK_PIN
LRC / WSEL	D26	GPIO 26	I2S_LRC_PIN
DIN	D25	GPIO 25	I2S_DIN_PIN

6. Controls (Buttons & Potentiometer)

Control	Terminal	Connects To	Purpose / Code Reference
Volume Pot	Pin 1 (Left outer leg)	ESP32 GND	Minimum volume reference
Volume Pot	Pin 2 (Middle leg / Wiper)	ESP32 D34 (GPIO 34)	VOLUME_POT_PIN
Volume Pot	Pin 3 (Right outer leg)	ESP32 3V3	MUST use 3.3V, NOT 5V
WiFi Button	Leg 1	ESP32 D32 (GPIO 32)	WIFI_BUTTON_PIN (internal pull-up)
WiFi Button	Leg 2	ESP32 GND	Shorts to ground when pressed
Mode Switch	Center Pin	ESP32 RX2 (GPIO 16)	MODE_SWITCH_PIN (Disc / BT mode)
Mode Switch	Outer Pin	ESP32 GND	Ground when switched

■ The Volume Pot's Pin 3 must use 3V3, not 5V — 5V can damage the ADC input.

7. LED Wiring

■ Rule: Never place two LEDs in parallel sharing a single resistor — one will always be brighter. Always give each LED its own resistor.

7.1 — WiFi Status LED (1 LED)

Leg	Connection
Anode (+)	220Ω or 330Ω resistor → ESP32 D33 (GPIO 33)
Cathode (–)	ESP32 GND

7.2 — Bluetooth Status LED (1 LED)

Leg	Connection
Anode (+)	220Ω or 330Ω resistor → ESP32 TX2 (GPIO 17)
Cathode (–)	ESP32 GND

Driven by firmware: BT_LED_PIN (GPIO 17 / TX2).

7.3 — System Status / Error LEDs (2x Red LEDs, parallel, own resistors)

LED	Leg	Connection
LED 1	Anode (+)	220Ω resistor → ESP32 D4 (GPIO 4)
LED 2	Anode (+)	220Ω resistor → ESP32 D4 (GPIO 4)
Both	Cathode (–)	Soldered together → ESP32 GND

Driven by firmware: RED_LED_PIN (D4 / GPIO 4) goes HIGH when flashing status/errors.

7.4 — Continuously-ON Power LEDs (2 LEDs)

LED	Leg	Connection
LED 1	Anode (+)	330Ω resistor → ESP32 3V3
LED 2	Anode (+)	330Ω resistor → ESP32 3V3

LED	Leg	Connection
Both	Cathode (–)	Soldered together → ESP32 GND

Using 3V3 with a 330Ω resistor gives a soft, battery-friendly glow. LEDs turn on/off with the power switch.

7.5 — External Charger Status LEDs (from TP4056, internal LEDs kept intact)

LED	Leg	Connection
Red (Charging)	Anode (+)	1kΩ resistor → TP4056 '+' input pad (near USB)
Red (Charging)	Cathode (–)	TP4056 Leg 2 (top row, 2nd pin from left)
Blue (Full)	Anode (+)	1kΩ resistor → TP4056 '+' input pad (near USB)
Blue (Full)	Cathode (–)	TP4056 Leg 3 (top row, 3rd pin from left)

8. Pins to Leave Disconnected (Do NOT Wire)

Several modules in this build have extra pins that are unused by design. Leaving them floating (no wire, no solder) is correct and expected — do not ground them or tie them to power unless noted otherwise.

Component	Pin(s) to Leave Disconnected	Why
Power Slide Switch	Pin 3 (the outer leg opposite the one wired to TP4056 OUT+)	Only the Center (common) and one outer leg are used to make an ON/OFF switch. The far outer leg has no function.
Mode Switch	The outer leg NOT wired to GND (i.e. whichever outer pin you didn't use)	Same 3-pin slide-switch logic — Center + one outer pin is all a simple two-position switch needs.
WiFi Button (if 4-leg tactile type)	The second pair of legs (the two legs diagonally opposite the ones you soldered)	4-leg tactile buttons have two pairs of legs that are internally shorted in pairs. You only need one leg from each pair — the duplicate pair can stay unsoldered.
RFID-RC522 Module	IRQ pin	IRQ is only needed if you implement hardware wake-up/interrupt-driven card detection. This firmware polls the reader instead, so IRQ is unused — leave it floating.
I2S Amp — MAX98357A	GAIN pin and SD (Shutdown) pin	GAIN floating = default +9dB gain (fine for most builds). SD floating = amp stays enabled via its internal pull-up. Only wire these if you specifically want to change gain or add a hardware mute/shutdown line.
I2S DAC — PCM5102 (if used instead of MAX98357A)	FLT, DEMP, XSMT, FMT, SCK	Most PCM5102 breakout boards already pull these to the correct level with onboard resistors. Check your specific board's silkscreen before wiring — on most versions these are pre-set and should not be touched.
TP4056 Module	NTC / TP pad (if present on your board)	This pad is for an optional battery thermistor. If your battery pack doesn't have one, leave this pad unconnected — it is not required for normal charging.

■ ESP32 Strapping Pin Warning: D12 (GPIO 12) is a strapping pin that affects the board's flash voltage setting at boot. It is intentionally unused in this project — do not connect any switch, LED, or sensor to D12, as an unexpected HIGH signal on it during boot can prevent the ESP32 from starting correctly.

Other ESP32 pins not used anywhere in this build

D15, D2, TX2, RX0, TX0, EN, VN, VP — these are simply not part of this circuit. You can leave them completely unconnected; they're free for future expansion (extra buttons, sensors, LEDs, etc.) and don't need to be grounded or wired to anything for the current design to work.

9. Pre-Solder Safety Checklist

- ■ Set MT3608 output to 5.0V with a multimeter BEFORE wiring it to anything.
- ■ RC522 RFID module → 3V3 ONLY. Never 5V.
- ■ Volume potentiometer's outer legs → GND and 3V3 (not 5V).
- ■ Every LED gets its own resistor — never share one resistor across two LEDs in parallel.
- ■ Double-check polarity (long leg = anode/+) on every LED before soldering.
- ■ Common ground: ESP32 GND, MT3608 VOUT–, SD module GND, RFID GND, and amp GND must all tie together.
- ■ Leave RC522's IRQ, the amp's GAIN/SD, and the unused switch legs floating — don't ground or power them.
- ■ Do NOT wire anything to D12 (GPIO 12) — it's a boot strapping pin.